Sina Shiri

Electrical Engineering Graduate | Embedded Systems & Research

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GitHub: sinashiry | Research and academic profile

Academic profile

Electrical engineering graduate seeking master's study in embedded computing and hardware/software systems. Built a practical research foundation through a Zynq-focused undergraduate thesis, C++/Python vision prototypes and technical book authorship. Combines industrial development at IstaSanat with current independent work in hardware security research.

Research interests

Embedded computing; hardware/software integration; computer vision on embedded platforms; intelligent sensing; hardware-security research.

Education

B.Sc. Electrical Engineering - Electronic Technology 2013–2016

Mohajer Technical University of Isfahan, Isfahan, Iran
GPA: **18.65/20** (original grading scale).
Thesis: "Detecting and Tracking Special Moving Objects in Public Environments on the ZYNQ platform."

Associate Degree in Electrical Engineering 2011–2013

Tabriz Technical and Vocational University, Tabriz, Iran
GPA: **17.85/20** (original grading scale).
Thesis: "Data Graph of Industrial Sensors in Noisy Environments without Distance Limit."

Technical books

- **S. Shiri** and H. Aalamshahi. *Full Reference of Raspberry Pi*, second edition. Elmiran, Tabriz, Iran, 2018. Persian-language technical book.
- **S. Shiri** and H. Aalamshahi. *Full Reference of Raspberry Pi*, Volumes 1 and 2. Elmiran / Ebadi, Tabriz, Iran, 2017. Persian-language technical books.

Selected research and development

Undergraduate thesis: vision on Zynq B.Sc. research project

Algorithm prototyping, C++/MATLAB, Vivado HLS, Zynq

Explored object detection and tracking from algorithm prototypes through an HLS and Zynq implementation workflow. Archived demonstrations show Sobel filtering, target-colour search and an implementation timing comparison.

Aerial Object Detection & Tracking

Computer-vision prototype

C++, OpenCV, target selection and image processing

Developed desktop prototypes for vehicle-region detection and selected-target tracking in recorded footage. Demonstrations show adjustable processing controls and target-selection modes.

Fire & Smoke Detection

Computer-vision / ML prototype

Computer vision, neural-network prototyping, embedded camera integration

- Explored image-processing and neural-network approaches to visual fire and smoke detection.
- Worked on dataset preparation, camera integration and a monitoring interface with event handling and status display.

Industrial Alarm Monitoring System

Embedded-system integration

Python, Raspberry Pi, industrial monitoring, system integration

- Developed monitoring software for 40 alarm inputs, integrating custom electronics with operator interfaces and database-backed event records.
- Implemented standalone and local-network workflows with configurable input modes, labels and controls.

Professional experience

Python Programmer & Hardware Designer

2017–Present

IstaSanat Engineering Company
Developed industrial control and monitoring software and integrated hardware, user interfaces and event logging.

Freelance Python Developer & Hardware Designer

Current

Currently working independently in hardware security research. Earlier freelance projects include caller-identification and attendance applications, hardware integration and Raspberry Pi technical writing.

Research and engineering methods

Programming: Python and C++; operator interfaces, client/server applications and data workflows.
Embedded and vision: Hardware/software integration, embedded Linux, device communication, object detection and tracking.
Hardware and design: PCB development; MATLAB, Altium Designer; experience with Zynq, Vivado, Verilog and VHDL.

Honors & distinctions

- **2016 – AUTCUP, third place:** Tele-Operated Deminer, team Red 2; Amirkabir University of Technology, Tehran.
- **2016 – AUTCUP, MAV participation:** Aras Robotics team; documented by the competition certificate.
- **2009–2010 – Provincial electronics recognition:** Recognised in the Scientific-Practical Competition and selected for the national stage.

Languages

Azerbaijani: native; Persian: bilingual; Turkish: full professional proficiency; English: professional proficiency. Levels are self-assessed.

Academic and professional references available on request.